

An End-to-End Pipeline for Scenario-Based Transportation Modelling under Deep Uncertainty

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Jan A. Ghadamian¹, Orlando M. Roman¹, Milos Balac², Bryan T. Adey¹

¹ETH Zürich, IBI-IM

²ETH Zürich, CSFM

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ABSTRACT

Future mobility planning requires tools that are both technically rigorous and accessible to a broad range of practitioners. This work presents an end-to-end pipeline that integrates scenario-based thinking with the agent-based modelling framework MATSim, via the synthetic population and trip generation tool Eqasim. The pipeline enables users to translate qualitative scenario drivers, such as demographic change, economic conditions, or technological adoption, directly into full simulation outputs, without manual intervention between pipeline stages. A systematic classification of fourteen scenario drivers by modelling feasibility provides a structured basis for future scenario construction. The possibility to include a graphical user interface (GUI) lowers the technical barrier, allowing non-specialist audiences to engage with the pipeline. The result is a reproducible, traceable, and extensible workflow for generating and comparing a wide range of plausible future mobility scenarios. To demonstrate the practical usability of the pipeline, a case study was conducted for the island of Corsica, France.

KEYWORDS: MATSim, Eqasim, Agent-Based Modelling, Scenario Planning, Future Mobility, Deep Uncertainty

1 Introduction

Infrastructure management in the mobility domain occupies a unique position within the strategic planning sciences: planners are required to deliver quantitatively robust results for an inherently uncertain future, in a context that can neither be fully modelled nor precisely predicted. This tension is often summarised more bluntly as: “*Transportation modelling is an attempt to predict something that will inevitably be wrong.*” Nevertheless, the value of transportation models lies not in predictive accuracy, but in their ability to systematically explore plausible futures and support evidence-based decision-making.

Contemporary challenges such as demographic shifts, technological disruption, and the climate transition, place unprecedented demands on transportation planning. Infrastructure decisions made today will shape mobility systems for decades. Planning authorities, public transport operators, and other policymakers therefore require robust methodologies for generating and comparing future scenarios that are both internally consistent and directly usable in simulation environments. This work presents a structured end-to-end pipeline that addresses this need, leveraging the MATSim framework alongside Eqasim to translate high-level scenario assumptions into simulated model outputs.

The challenge of generating realistic synthetic populations has been widely acknowledged in the literature. The present work builds largely on the methodology of Hörl and Balac [2], who propose a reproducible framework for constructing synthetic populations and demand from publicly available data in France [1]. The importance of translating qualitative drivers into model inputs has likewise been recognised: Rasca [3], for instance, integrates qualitative interview data into agent-based models, though with the aim of improving current-state modelling accuracy rather than exploring future scenarios.

This is precisely the gap the pipeline presented here aims to fill. No established framework currently provides a structured, reproducible pathway from high-level scenario assumptions to simulation-ready agent-based **future** model inputs. The present work addresses this by combining a systematic driver classification with an automated Eqasim–MATSim pipeline, enabling the direct and traceable translation of future scenario definitions into quantitative mobility simulations.

2 Methodology

2.1 Scenario Drivers and Modelling Feasibility

The first step in the pipeline is the identification and formalisation of scenario drivers: the key dimensions along which future conditions may vary. Through a structured expert process conducted with the Swiss Federal Railway (SBB), fourteen drivers were identified (see Table 1).

A central contribution of this work is a four-level classification framework that categorises each driver according to the effort required for its integration into the Eqasim–MATSim pipeline. Level A drivers translate directly into model parameters with a rather small configuration effort, such as modification of the utility functions. Level B drivers are tractable but require more substantive model development, such as dedicated MATSim extensions. Level C drivers are multi-dimensional and demand the integration of heterogeneous data sources and coordinated modelling decisions. Level D drivers cannot be meaningfully represented within the daily-activity simulation paradigm of MATSim, as they operate at timescales or levels of disruption that fall outside the framework’s equilibrium-seeking assumptions.

Level	Driver	Integration Approach
A – Direct	Economy, Behaviour, Living, Population, Labour	Configuration-level adjustments to cost functions, utility parameters, and synthetic population or activity attributes
B – Moderate	Technology, Wealth, Journeys, Leisure, Time	Dedicated MATSim extensions or extended trip generation logic; requires careful validation
C – Complex	Regulations, Energy, Goods	Multi-dimensional; requires heterogeneous data sources and coordinated implementation across pipeline stages
D – Unfeasible	Climate, Security	Outside the scope of daily activity-based simulation; incompatible timescales or singular disruptive nature

Table 1: Classification of scenario drivers by modelling feasibility

2.2 End-to-End Pipeline and Interventions

Beyond the seemingly uncontrollable scenario drivers described above, the pipeline also accommodates direct network-level interventions: Extensions or reductions of the road network capacities and/or freespeeds, the addition or removal of public transport services, and modifications to active mobility infrastructure.

The pipeline is designed to be modular: Users can specify a desired scenario by selecting specific drivers and add them to the pipeline at the right stage. The pipeline then propagates these choices automatically through the Eqasim synthetic population and trip generation stages, before passing the resulting inputs to MATSim for simulation. Outputs conform to the standard MATSim format, enabling the full suite of existing analysis and visualisation tools to be applied without modification.

By encapsulating the translation logic between scenario assumptions and model parameters, the pipeline ensures that scenario construction is reproducible. Each scenario is fully defined by its input configuration and the random seed, enabling straightforward comparison across scenario sets and facilitating sensitivity analysis.

3 Case Study: Corsica

To demonstrate the practical applicability of the proposed pipeline, a case study was conducted for the island of Corsica, France. Corsica represents a well-suited testbed for scenario exploration for three key reasons. First, it is a geographically self-contained territory with a clearly defined transport network. Second, its population size is moderate, allowing simulations to be performed within reasonable computational limits. Third, comprehensive public data provided by the French government are readily available and can be directly integrated into the Eqasim pipeline, as described in [2].

Three categories of modifications were applied to the baseline model:

1. **Population driver:** This component was used to simulate changes in overall population size. The pipeline generates a rescaled synthetic population while preserving the underlying demographic structure. Optionally this underlying demographic structure can even be further modified through user-inputs.
2. **Behaviour driver:** Changes in modal preferences were introduced by adjusting the alternative-specific constants (ASC) within the mode choice model. This allows the representation of populations with differing preferences towards available transport modes.
3. **Network interventions:** Direct modifications to the transport network were implemented, including changes to road network topology, in order to reflect plausible infrastructure investment or restriction scenarios.

The case study highlights the end-to-end nature of the pipeline. Starting from a limited set of high-level scenario parameters defined by the user, the system automatically generates a modified synthetic population, updates trip generation inputs, and executes a complete MATSim simulation, producing standard output files. The results indicate that even relatively modest changes in population size, behavioural preferences, and network supply lead to clearly distinguishable outcomes in terms of modal split and network load. This demonstrates the sensitivity and responsiveness of the pipeline to scenario inputs.

4 Contributions and Outlook

This work makes the following key contributions:

1. **Empirical validation:** The Corsica case study provides a concrete demonstration of the pipeline's usability and sensitivity, validating its applicability within a real-world context.
2. **Scenario generation framework:** The proposed end-to-end pipeline enables the rapid and consistent generation of diverse future scenarios within the MATSim ecosystem. The resulting dataset forms a foundation for subsequent research, particularly in the area of surrogate modelling for network-based simulators.

Future work will focus on extending the range of incorporated future drivers, particularly across levels A–C (see Table 1), thereby broadening the scope of scenario-generated synthetic populations. This extension will enable more systematic scenario comparison studies. Furthermore, the pipeline will be released as an open-source tool to support and foster further research within the MATSim community.

References

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